



Ajay Kumar Tiwari

Solution Architect | AI/ML, Gen AI

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PROFESSIONAL EXPERIENCE

- 16 years of experience with Gen AI & RAG Solutions and Cloud Architecture with Hands-On approach towards solving business problems and creating value.
- Having Product Development Experience from Concept Phase to Design, Development, UAT and Support Handover.
- Leading Customer and Stakeholder discussions for developing **IT Products with insight into Technology Trends and Driving Business Value.**
- Experience in reporting, communication & influencing **Business Owners, Decision Executives & Senior Leadership** level stakeholders.
- Technology Leadership experience in **Industry 4.0, Digital, Data, Analytics & Visualization, Cloud, Gen AI/ AI/ML based solutions** for enterprise programs.
- **Digital Twin and IoT Implementation for Oil & Gas domain.**
- Implementation of **Agile** (Scrum, Kanban), **DevSecOps**, CI/CD & Traditional **Waterfall** SDLC delivery, ITIL, SCRUM.
- **Certified Scrum Master (CSM)**, with extensive experience in 10+ years' experience in Agile Scrum ○

Microsoft Azure Solution Architecture Exposure / Azure Well Architected Framework

CERTIFICATIONS

- The Open Group - TOGAF 9.1 certified
- Azure AI Certification
- Azure Cloud Solution Architect
- NVIDIA Deep Learning Institute – Online Courses
 - Getting Started with Deep Learning/ Building Transformer Based Natural Language Processing
- Completed AI Engineer Master Program (6 months / Simplilearn)

SKILLS/Knowledge

Technology Skills	Leadership Skills
C++.NET Core, Python, PyTorch, C++, PySpark	Stakeholder Engagement
Azure Cloud , AWS, Google Vertex	Storytelling Skills
Gen AI use cases using Databricks – Data Engineering, mlflow, MLOps	Influence/ Decision Making
LLM, Small Language Model, Huggingface Models	Problem-Solving, Critical Thinking
Langchain, RAG Architecture, LangGraph	Coaching & Mentoring
Solution Architecture High-Level Design / IT Landscape Design	Agility
Architecture Tools – Visio, Bizz Design	Stakeholder Engagement
Technical Architecture	
Integrating Solutions	

Logic Apps, MuleSoft, API Management, Integration Patterns		
SAP ERP (Plant Maintenance Module) – Tcode, RFC, Tables, Functional Location, Equipment etc. / SAP Integration with Cloud based SaaS		
AI / ML / NLP – Azure API , OpenAI APIs		
Agile SCRUM/Azure DevSecOps/CI CD/GitHub/AWS DevOps		
Software Engineering		
Azure managed Containers/Microservices/ container orchestration systems /Cloud Native /Docker, Kubernetes		
C# .Net technologies, C#, .Net Core, MVC, Web API JavaScript, Flutter (used in Solution Design)		
MS SQL Server		
Requirement Gathering, User Story Creation, Sprint Planning, Backlog Grooming, Sprint Demo		
IaC – Terraform, Puppet, Chef, Ansible, Jenkins		
Open Source Libraries (C++ & Python), Linux		
Knowledge of Google GCP		
Front End – React, JavaScript		

INDUSTRY

- Oil & Gas – Manufacturing Chemical & Oil Products
- Asset Information Management – Oil & Gas
- Digital Twin Implementation for Oil & Gas Industry, IoT Implementations
- Revenue Cycle Management (RCM)

WORK HISTORY

Solution Architect

Expleo Solutions Limited (July 2023 – till Date)

Provide Architecture roadmap and Architecture Ownership and collaboration with Technical Teams. Designing Solutions for Application Architecture, Application Modernization, Data and analytics Platforms, Data Engineering Use cases on Azure Cloud, AI/ML, Gen AI use cases.

Currently working on Gen AI Platform Architecture design for Banking & Finance Domain client Project - enables enterprises to seamlessly adopt secure, enterprise-grade AI solutions, prioritizing data privacy, robust governance, and operational efficiency. This Platform is used for AI/ML, NLP, Gen AI use case development. Aspects related to Data Privacy, Compliance, Enterprise Data Policies are handled based on Non-Functional Requirements. Experience with Traditional Deep Learning based Computer Vision Models, BERT, GPT, ML Models and MLOps, Python, PyTorch, Langchain, RAG, Agentic AI, LangGraph

Use cases include Application Modernization Projects which include Rearchitecting Applications from Monolithic to Microservices Architecture and Setting up CI CD using Azure DevOps Pipelines, Microservices Architecture, React, IaC.

Solution Architect

Shell India Markets Private Limited (Nov 2015- Sep 2022)

Digital Twin for Oil & Gas Business

- Digital Twin implementation for Oil & Energy Domain, A digital twin is a virtual representation of a physical object or system and dynamic behaviour over time. Started with Pilot for few sites and later continued with Production

deployment. Involved with Technical Integration work, SaaS solution Integration with Shell On-premises data sources.

- Responsibilities – High Level Solution Design, Technical Architecture work for 3D Visualization Solution and Rollout of Unified Cloud Engineering (PaaS App) making sure aligned with Architecture standards, API Development using Azure Logic Apps & MuleSoft, Industrial IOT, Streaming Data sources, Gen AI Use cases, Requirement Gathering, Gap analysis for Data Architecture - integrating On-premise data source with Cloud Solution. Presenting Solution in Architecture Review Board. Azure Synapse, Azure Data Lake, Azure Data Factory Pipeline setup, Databricks, PySpark, AI/ML, BERT, LLM, GPT, Gen AI Python, Langchain, RAG, Agentic AI LangGraph. Scala used for Data Processing, Azure Functions C++, C# .NET, Python, PyTorch, CI CD using Azure DevOps Pipelines, IaC using Terraform.

Technical Analyst

Asset Data Platform – Supply Business Shell

- Downstream Supply – Data Platform Shell Downstream Supply Business required Visualization of Supply Terminal data for meaningful insight and better business decisions. Currently (As-Is) Data is scattered in multiple systems, causing delays in decision making. Project was setup to deliver scalable, near real-time data Analytics Platform using Azure Cloud Offering – Azure Data Factory, Azure Data Lake, Azure Data Bricks, Synapse data warehouse and Power BI.

Responsibilities – High Level Solution Design, Technical Architecture and Data Integration design. Op-premise data source integration with Azure Cloud. Supported Power BI Dashboard and near real time Use cases to stream data from IoT devices. Solution Design Review in Architecture Review Board, IT Landscape Design and New requirement solution design and Optimize Azure cost. Azure Data Factory Pipeline setup, PySpark, AI/ML, Deep Learning, Python, Azure Functions, C++, C#. NET, CI/CD using Azure DevOps Pipelines, Terraform, IaC.

Technical Analyst

Asset Information Management (AIM)

- Development of Shell Global Asset tracking system. Asset Information Management is a distributed enterprise system that integrates data stored in SAP R3 with data stored in Oracle 10 and electronic documents stored in IBM FileNet P8 and is in use for about 15+ years within the Shell organization with thousands of users. Tools and technologies used: Visual Studio 2010, C++ with MFC, NET/C# and C++, ASP.NET, Progress Orbix CORBA 3.3 middleware, SAP R3 Remote Function Calls, Oracle Pro/C. The development process is SCRUM based, Python and C#.NET, CI , CD using Azure DevOps Pipelines, Terraform, Microservices, IT Landscape Design.
- Main Responsibilities: Implement Solution Design, and new functionality in the system and improve overall quality and stability. Worked for Multiple releases, Build and Release processes.

Consultant

Capgemini India Private Limited

(Jan 2012- Nov 2015)

- Worked with Revenue Cycle Management Product, CBO application, calculates Expected Reimbursement amounts for Hospitals and USA Healthcare providers.
- Worked as Developer for product Contract Manager and roles include New Feature Enhancement and Defect Fix these items were prioritized by Business Analysts. Writing Unit Test Cases, Code Reviews Worked in Agile Methodology having Daily standup calls and iteration sizing and story time for items prioritized by Business Analysts, deployment for DB. Working in a Continuous Integration manner CI/CD Pipelines using DevOps of Agile and Customer reported issues are worked in a KANBAN way and Items going to be worked in Iterations are discussed in Agile meetings Storytime and sizing.
Technology used is C++/VC++/STL / C# .NET / Azure DevOps UML and Database on SQL Server 2008/10 and Smart Pointers were used to prevent memory leaks (shared pointers) and multithreading.

Senior Software Engineer

Unisys Global Services Limited (Oct 2008– Dec 2009) ○ Agile Business Suite (Product Development). This is a modelling tool based on UML. 4GL programming languages LDL+ used for Programming Worked on issues reported by customers and New Enhancements, development work (Insurance and Finance Domain). Technology used is C++ with STL/WIN32SDK.with VS2005

Developer

ABB Global Services Limited (Dec 2005 – Sep 2008) ○ IEC 61850 is a Communication Protocol used in Substations. Here “Connect” – means deploying new Object Types to 800xA. My role is more into the Retrieve component that parses. SCD file (XML file) and maps to. OCS file format (ABBspecific file format) and, I worked on the Retrieve Component, Industrial IoT.

- Technology used is C++

EDUCATION

- Bachelor of Engineering (Information Technology) – 2001-2005 Rajiv Gandhi Proudhyogiki Vishwavidyalaya

ABOUT ME

I enjoy running and cycling.